

Project Management Protocol of the Netherlands eScience Center

Authors: Netherlands eScience Center

Date: July 2026

Version: 4.0

License: CC BY 4.0

Date	Version	Changes
1 July 2026	4.0	Major changes, renew protocol after 2025 restructuring.
1 June 2025	3.5	redacted internal references, added RSD checklist appendix and missing intro text, included Zenodo references, fixed tables, and improved description of annual review meeting
1 December 2024	3.0	Planning, Legal agreements, Project budget breakdown, change in the workshops and end report processes, open-access publications
1 October 2023	2.0	DT and PM mandates, Roles of Directors, PM role in Ambition 2, SS, KD and external projects, Communications role, Editorial team, breakdown of hours, opportunities beyond project, Technology Status Report, End report
23 September 2022	1.0	initial version

Contents

1	Scope and definitions	3
1.1	Scope	3
1.2	Types of projects	3
1.3	Stakeholders	4
1.4	Project lifecycle overview	6
2	Project Initiation	7
2.1	SH assignment	7
2.2	Project registration	7
2.3	Project staffing	8
2.3.1	Staffing preparation	8
2.3.2	Team assignment	8
2.3.3	Project planning	8
2.4	Administrative kick-off meeting	9
2.5	Kick-off meeting	10
2.6	Legal agreements	11
2.7	Technology plan	11
3	Project execution	14
3.1	Project development	14
3.1.1	Data, infrastructure and computing resources	14
3.2	Project meetings	15
3.2.1	Regular project team meetings	15
3.2.2	Internal coordination and escalation	15
3.2.3	Workshops and community events	16
3.3	Project monitoring	16
3.3.1	Project log	17
3.3.2	Progress, hours and budget monitoring	17
3.3.3	Risks, issues and project changes	18
3.4	Project quality and outputs	19
3.4.1	Software quality and sustainability	19
3.4.2	Data and Software Management Plans	19
3.4.3	Project documentation	20
3.4.4	Project outputs	20
3.5	Knowledge transfer and communication	21
3.6	Project review	21

3.6.1	Phase 1 – Initiating the review cycle	22
3.6.2	Phase 2 – Internal project assesment	22
3.6.3	Phase 3 – Review meeting	22
3.6.4	Phase 4 – Continuation decision	23
3.6.5	Phase 5 – Financial processing and communication	24
4	Project closing	25
4.1	Closing meeting	25
4.2	End report	25
	Appendices	29
A	Applying the protocol to external projects	29
A.1	Project initiation	29
A.2	Project execution	29
A.3	Project closing	30
B	Applying the protocol to internal projects	31
C	Lead RSE role description	32
D	Project lifecycle checklist	33
E	RSD project pages	34
F	Example of the project log	36
	Key eScience Roles appearance	37

1 Scope and definitions

1.1 Scope

This document is the official project management protocol of the Netherlands eScience Center. It describes the activities, responsibilities, and procedures required to manage an eScience project after it has been formally awarded, through to its formal closure.

The protocol primarily applies to projects awarded through the eScience Center calls for proposals, although it also provides guidance for other types of projects as defined in Section 1.2. It follows the project life cycle and, for each phase, specifies the required activities, optional activities, responsible roles, and expected outcomes.

The call process itself is governed by a separate protocol [1] and is therefore outside the scope of this document. This protocol starts once a project has been formally awarded and covers all activities until the project is formally closed.

The following topics are also outside the scope of this protocol:

- independent project evaluation (impact, outputs, process, and collaboration);
- detailed guidance on project management tools (e.g. [Gantt](#), [Metabase dashboards](#), [AFAS](#));
- general project management methodologies and techniques.

The Management Team (MT) approves this protocol and reviews it annually. The document is organised according to the project life cycle (Section 1.4), presenting activities in chronological order.

1.2 Types of projects

The eScience Center receives an annual budget from [NWO](#) and [SURF](#), the larger part of which is allocated to projects submitted by researchers working at eligible research performing organizations in the Netherlands in the form of the in-kind provision of Research Software Engineers (RSEs). The eScience Center also participates in externally funded projects and carries out internally funded strategic initiatives.

These projects differ in their funding model and governance. Throughout this protocol, they are grouped into the categories shown in Table 1.

Table 1: Project categories covered by this protocol

Project type	Description	Governed by
Call projects	Projects funded through the Netherlands eScience Center calls for proposals and delivered through in-kind RSE support.	This protocol
External projects	Projects funded by external organisations (e.g., NWO, Horizon Europe, industry). External contractual obligations take precedence where applicable.	This protocol + external agreements. See Appendix A for further details.
Other projects	Internal initiatives such as Research and Development (R&D) projects, Fellowship projects, and Dissemination and Community (D&C) initiatives.	This protocol where applicable + dedicated internal procedures. See Appendix B for further details.

1.3 Stakeholders

An eScience project is a project involving the eScience Center, where responsibility is shared between different stakeholders who each have their own roles and responsibilities during specific phases of the project.

Table 2: Project stakeholders and their roles

Stakeholder	Role in the project	Further information
Lead Applicant (LA)	Represents the applicant organisation and is the primary external contact for the project. Accountable for the scientific contribution and the participation of the applicant project team.	Responsibilities may also be defined by the call conditions, grant agreement, or consortium agreement.
Lead Research Software Engineer (Lead RSE)	Coordinates the eScience Center's project activities, supports the successful delivery of the agreed work, and serves as the primary contact for the project.	See the formal Lead RSE role description in Appendix C.
Research Software Engineer, assigned to a project (RSE)	Contributes research software engineering expertise and carries out agreed project activities under the coordination of the Lead RSE .	See the RSE role and job descriptions [2, 3].
Consulting Research Software Engineer (Consulting RSE)	Provides specific expertise to support concrete technical aspects of a project.	Works under the coordination of the Lead RSE and is involved when additional expertise is required.
Section Head (SH)	Provides line management for the eScience Center project team, ensures appropriate staffing and capacity, and is accountable for the successful delivery of the eScience Center's contribution to the project.	Assigns the Lead RSE and project RSEs, and oversees team capacity and personnel matters. See Section 2.3.
Programme Officer (PO)	Coordinates the call, review, and project selection process, and supports the administrative management of the programme throughout the project lifecycle.	The funding decision communicated by the Programme Officer marks the starting point of this protocol.
Secretary	Provides administrative support for project coordination and formal meetings.	Assists with meeting organisation, documentation, and project communications.
Policy & Compliance Officer (P&CO)	Provides guidance on legal, policy, and compliance matters related to the project, including agreements with external partners.	Supports the preparation and review of collaboration agreements, NDAs, licences, and other legal documents where required.
Technology Officer (TO)	Provides strategic advice on technology and software engineering.	Supports the Center's technology strategy, technical expertise, and R&D programme.

Continued on next page

Table 2: Project stakeholders and their roles (Continued)

Stakeholder	Role in the project	Further information
Community Manager (CM)	Supports community building, engagement, and outreach activities related to the project.	Provides advice on developing community engagement plans and connecting projects with relevant research communities.
Training Programme Coordinator (TPC)	Supports the planning and delivery of the digital skills training programme.	Advises the project team on available training programmes, coordinates project-specific training where appropriate, and supports the development of training materials and workshops.
Communications	Supports the communication and dissemination of project activities and outcomes.	Advises on communication strategies and promotes project achievements through appropriate communication channels.
Finance	Responsible for project financial administration, budget monitoring, financial reporting, and project closure activities.	Comprised of the Financial Controller and the Financial Administrator.
Performance Manager (PerfM)	Oversees portfolio performance, capacity planning, and portfolio reporting using project information and financial data. Supports MT decision-making on resource allocation and annual budgeting.	Operates at portfolio level rather than individual project level.
Management Team (MT)	Provides strategic oversight and approves project decisions that fall outside the delegated authority of the SH .	Includes the General Director, Section Heads, Portfolio Manager, and Performance Manager. The General Director is the final decision-maker.
eScience Center project team	The team responsible for delivering the eScience Center's contribution to the project.	Comprised the Lead RSE , RSEs, Section Head, and other contributors involved on behalf of the eScience Center.
Project team	The complete team responsible for delivering the project.	Comprised the eScience Center project team together with the Lead Applicant and the collaborating project partners.

1.4 Project lifecycle overview



Figure 1: Project Lifecycle Overview

At the eScience Center, a standard project lifecycle consists of three phases (see Figure 1) initiated, followed by the execution phase, during which research software is developed and project activities are coordinated and monitored. Finally, the project is formally closed through a structured closing process.

The following sections describe each phase of the project lifecycle in detail. A summary checklist of the key activities and deliverables for each phase is provided in Appendix D.

2 Project Initiation

The project initiation phase begins once a project has been approved for funding. Following the communication of the awarded project by the Program Officer (PO), the responsible **SH** initiates the project setup. The purpose of this phase is to establish the project within the eScience Center by completing the required administrative setup, assigning responsibilities, and preparing the project for execution.

The exact initiation process depends on the project type. External projects, as well as certain internal projects, may require additional contractual or legal arrangements before they can commence. Further information on the different project types is provided in Appendix B and Appendix A.

Once the project has been registered, staffed, and all applicable administrative, financial, and legal requirements have been fulfilled, it is ready to enter the execution phase.

2.1 SH assignment

Each project has one assigned **SH**, who is accountable for the project throughout its lifecycle.

For projects funded through the eScience Center's regular funding calls, the assigned **SH** normally corresponds to the section under which the proposal was submitted. In other cases, or where the assignment is not self-evident, the Section Heads jointly determine the most appropriate assignment, taking into account the project's scope, expertise, and organizational responsibilities.

Once the assignment has been agreed, the responsible **SH** is recorded and **Finance** is notified to initiate the project registration process.

Responsible: Section Heads.

2.2 Project registration

The table below summarizes the main project registration outputs, the responsible stakeholder, and the associated activities.

Table 3: Project registration activities and responsibilities

Output	Responsible	Main actions
Project record	Finance	• Creating the project code in AFAS • Assigning the responsible SH as budget holder
Project portfolio	Finance	• Creating the project portfolio folder [4] following the required template sub-folders • Requesting missing project documentation • Uploading the grant package (proposal, awarding letter, signed start form, and supporting documentation submitted by the LA)

Responsible: Finance.

2.3 Project staffing

The responsible **SH** is responsible for initially staffing the project, including the appointment of a **Lead RSE** and the assembly of the eScience Center project team [5]. Staffing decisions take into account the project's requirements, available capacity, and the expertise required for successful delivery. Once the project has started, the **Lead RSE** may adjust the staffing as needed (see Section 3.3.3). The **Lead RSE** will also serve as the primary point of contact for the LA.

2.3.1 Staffing preparation

The responsible **SH** reviews the project proposal to determine the staffing requirements for the project. The assessment considers the project scope and objectives, the awarded budget, the expected timeline, the required scientific and technical expertise (making use of available organizational information on technology capabilities, e.g., technology skills surveys), any applicable contractual, legal, or organizational constraints, and the available capacity within the section based on the organizational planning system. These considerations form the basis for assigning the Science Center project team.

2.3.2 Team assignment

Every project must have: one accountable **Lead RSE**; and at least one additional Research Software Engineer assigned to the project. The eScience Center project team is assigned according to the following process:

1. The responsible **SH** identifies a suitable candidate for the **Lead RSE** role, taking into account the staffing preparation described above.
2. The **SH** discusses the proposed assignment with the candidate.
3. The proposed staffing is reviewed and confirmed by the Section Heads.
4. The responsible **SH** formally assigns the **Lead RSE** and the remaining project team members.

Once the **Lead RSE** has been assigned, they create the project page in the Research Software Directory (**RSD**, see also Appendix E). The RSD page serves as the central location for project information, including project description, team members, software outputs, publications, and other research products. The page should be kept up to date throughout the project lifecycle by the **Lead RSE**.

2.3.3 Project planning

Following project registration, the project is synchronized to the organization's planning system. The responsible **SH** records the **Lead RSE** in the corresponding project in Ganttlic. The **Lead RSE** prepares the initial project planning by allocating the assigned team members across the different project phases. The planning serves as the baseline for project execution and may be updated during the project as required.

Responsible: SH.

2.4 Administrative kick-off meeting

Once the project has been registered and staffed, the responsible **SH** organizes an administrative kick-off meeting with the LA. Prior to the meeting, the **SH** and **Lead RSE** review the project proposal and identify any project-specific requirements that should be discussed with the LA. The purpose of the meeting is to introduce the collaboration with the Netherlands eScience Center and to ensure that the project is administratively and organizationally ready to start. These may include:

- any additional organizational or technical support required for the project;
- deviations from the default intellectual property or software licensing arrangements;
- data readiness, data access requirements, and any personal data or GDPR requirements;
- software sustainability expectations; and
- ethical, legal, or other project-specific risks.

The meeting is supported by a standardized administrative kick-off presentation available in the project portfolio under the *Templates projects* folder [4]. The presentation is periodically updated to reflect the requirements of different funding calls and the eScience Center's current procedures and policies. The responsible **SH** uses this presentation to ensure that all standard topics are covered consistently across projects while adapting the discussion to the specific needs of the project.

The presentation typically covers the following topics:

- Introduction to the Netherlands eScience Center (mission, organization, and expertise);
- Working with the eScience Center (roles and responsibilities, collaboration model, in-kind and in-cash contributions);
- Project lifecycle and project governance;
- Community activities and research impact (Research Software Directory, Digital Skills Programme, Fellowship Programme, etc.);
- Intellectual property, software licensing, publication practices, and open science principles;
- Project-specific needs, including scientific and technical expertise, SURF infrastructure requirements, and additional resources (e.g., AI subscriptions or other project-specific services).

The presentation used during the meeting and any meeting notes are stored in the corresponding project portfolio folder together with the other project documentation. The Secretary may assist the **SH** with scheduling the meeting.

Table 4: Administrative kick-off meeting overview

Timing:	Soon after project registration and staffing have been completed.
Participants:	SH (organizer), LA, Lead RSE (optional). Others may be invited as needed.
Objective:	Introduce the collaboration with the Netherlands eScience Center, discuss the project setup, and identify any project-specific administrative or organizational requirements before project execution begins.
Duration:	1 hour.
Location:	Online.

Responsible: **SH**.

2.5 Kick-off meeting

Following the administrative kick-off meeting, the **Lead RSE** organizes the project kick-off meeting with the project team, comprising the eScience project team, the LA, and the LA's team. The purpose of the meeting is to establish a shared understanding of the project objectives, work plan, responsibilities, and expected deliverables before project execution begins.

Prior to the meeting, and based on their review of the project proposal, the **Lead RSE** and the eScience Center project team identify any scientific or technical aspects that should be discussed with the LA. These may include:

- the scientific objectives and expected project outcomes;
- the proposed work packages, milestones, timeline, and overall feasibility;
- technical challenges, assumptions, implementation risks, and whether additional technical expertise or consultation from other RSEs is required;
- software sustainability considerations and the initial Software Management Plan (SMP);
- dependencies on data, infrastructure, or external collaborators; and
- any issues requiring clarification before project execution starts.

The meeting is supported by a standardized project kick-off agenda available in the project portfolio under the *Templates projects* folder [4]. The **Lead RSE** uses this agenda to ensure that all standard topics are covered consistently across projects, while adapting the discussion to the specific needs of the project.

The agenda typically covers the following topics:

- Round of introductions;
- Project introduction and scientific objectives;
- Review of the project work plan;
- Agreements on the initial project planning and deliverables;
- Software sustainability and the Software Management Plan (SMP);
- Any other business.

The meeting outcomes, agreed action items, and any presentation material are stored in the corresponding project portfolio folder. The **Lead RSE** updates the project planning accordingly and prepares the project for the execution phase. These documents establish the initial project baseline and provide the starting point for the project log maintained during the execution phase. The Secretary may assist the **Lead RSE** with scheduling the meeting.

Following the kick-off meeting, the **SH** and **Lead RSE** determine, together with the Communications team where appropriate, whether internal or external communication activities should be initiated. These may include announcing the project on the eScience Center website, preparing news items for the kick-off meeting, or coordinating other outreach activities.

Note: The **Lead RSE**, responsible **SH**, and LA are expected to attend the meeting in person at the eScience Center or the project location. Other project team members may participate online if appropriate.

Responsible: **Lead RSE.**

Table 5: Project kick-off meeting overview

Timing:	Shortly after the administrative kick-off meeting.
Participants:	Lead RSE (organizer), SH , LA, project team. Others may be invited as needed.
Objective:	Establish a shared understanding of the scientific objectives, work plan, responsibilities, and expected deliverables before project execution begins.
Output:	Action items agreed planning, initial project baseline.
Duration:	1.5 hours.
Location:	At the eScience Center or the project location.

2.6 Legal agreements

The eScience Center promotes open, reproducible science and open-source software development. Projects funded through the eScience Center are carried out under the eScience Center Terms and Conditions [6]. However, project partners may require additional legal agreements to grant access to their facilities, data, infrastructure, or other resources. These agreements may also contain provisions related to intellectual property, confidentiality, data protection, or information security.

RSEs must not sign any formal agreements without consulting the responsible **SH**. Such documents include, but are not limited to:

- Guest agreement
- Data sharing agreement
- Non-disclosure agreement
- Consortium agreement
- Collaboration agreement
- Intellectual property (IP) agreement

The **SH** reviews the proposed agreement, consulting the **Lead RSE** where project-specific input is required, and determines whether additional review is required. If needed, the **SH** consults the Policy & Compliance Officer (P&CO) before the agreement is signed. The final signed agreement is archived in the corresponding project in the Project portfolio (see [4]).

Responsible: **SH**.

2.7 Technology plan

Every eScience project involves technological choices that influence software development, data management, reproducibility, and long-term sustainability. Following the project kick-off meeting, the **Lead RSE** prepares a technology plan in collaboration with the Lead Applicant and the project team. The technology plan is stored in the corresponding project portfolio (see [4]) and provides a shared reference for project execution.

The technology plan typically describes:

- selected technologies and rationale;
- software architecture or implementation approach;
- expected software features and data outputs;
- quality standards;

- reproducibility and sustainability considerations;
- key technical assumptions or constraints.

The technology plan is maintained as a living document and may be updated as the project evolves to reflect significant technological decisions agreed between the **Lead RSE** and the Lead Applicant and project team. An example technology plan is available in [7].

The **Lead RSE** is encouraged to consult colleagues with relevant technical expertise when preparing the technology plan. Where appropriate, the Community Manager (CM) may advise on software dissemination, community engagement, and strategies to promote software reuse and adoption. Likewise, the eScience Center's Special Interest Groups (SIGs) may provide guidance on domain-specific technologies, software engineering practices, and other technical topics relevant to the project.

Table 6: Technology plan overview

Prepared by:	Lead RSE , in collaboration with the LA and the project team. Where appropriate, colleagues with relevant technical expertise, the Community Manager, and Special Interest Groups (SIGs) may provide advice.
Target audience:	Project team.
Timing:	Shortly after the project kick-off meeting and before substantial software development begins.
Objective:	Document the technological decisions agreed for the project, including the selected technologies, implementation approach, software quality considerations, and other technical decisions that guide project execution.
Format:	Free-format document, maintained as a living document throughout the project as needed.

Responsible: **Lead RSE**.

3 Project execution

Once the project has been initiated, the execution phase focuses on delivering the agreed research objectives while monitoring progress, resources, risks, and deliverables. This is typically the longest phase of an eScience project and involves close collaboration between the eScience Center project team and the LA's team.

The **Lead RSE** coordinates the day-to-day technical execution of the project and facilitates regular meetings with the project team and the LA. The **SH** oversees project delivery from the eScience Center's perspective and supports the project when administrative, staffing, financial, or organizational intervention is required. The **Lead RSE** monitors progress to ensure that technical work, milestones, and deliverables remain aligned with the project objectives.

Because research projects vary considerably in scope, size, and complexity, the eScience Center does not prescribe a single project management methodology ([8–13]). Project teams are encouraged to adopt the practices and communication approaches that best suit their collaboration.

3.1 Project development

Project development is a collaborative effort between the eScience Center project team and the LA's team to achieve the scientific objectives defined during project initiation. Throughout the execution phase, the **Lead RSE** coordinates the technical work, facilitates communication between all team members, and ensures that development activities remain aligned with the agreed project objectives and planning.

Research software development is typically carried out iteratively. Scientific discoveries, user feedback, and technical challenges may require adjustments to the implementation approach. The **Lead RSE** and project team are expected to make informed technical decisions throughout the project, balancing scientific objectives with software quality, maintainability, sustainability, and the efficient use of available resources. Significant changes to the technical approach should be reflected in the Technology Plan (Section 2.7) and communicated to the project team.

Software developed at the eScience Center should follow the software development guide and established community best practices [8, 14]. At the beginning of software development, the **Lead RSE** establishes the project public repositories and development workflow, ensuring that the project team has appropriate access and that the repositories are linked from the project's RSD page. Whenever appropriate, project repositories should be based on, and build upon existing resources available in the [Netherlands eScience Center GitHub](#) organization.

Responsible: **Lead RSE**.

3.1.1 Data, infrastructure and computing resources

Research projects may require access to computing infrastructure, data storage, cloud services, or specialized research facilities. The **Lead RSE** and the LA determine the project's infrastructure requirements during execution and ensure that appropriate resources are available throughout the project.

The Netherlands eScience Center primarily relies on the national research infrastructure provided by [SURF](#), which includes high-performance computing (HPC) systems, cloud services, and data storage facilities. Depending on the project, access to SURF services may be obtained through project-specific allocations, the LA's institution, or dedicated test accounts. In addition, the eScience Center provides access to the [DAS-6](#) distributed supercomputer.

Snellius test accounts can be requested by Lead RSEs through the [SURF Service Desk](#). Access to SURF Research Cloud, as well as requests for new infrastructure or additional computing resources, should be coordinated with the [eScience Center's SURF liaisons](#). They can advise on the most appropriate services and the current application procedures. Up-to-date instructions and contact information are maintained on the eScience Center Intranet [15].

Responsible: **Lead RSE (consults SURF liaisons where appropriate)**.

3.2 Project meetings

Project meetings provide the primary forum for coordinating scientific and technical work, discussing progress, resolving issues, and planning future activities. Different types of meetings serve different purposes throughout the execution phase and should be adapted to the needs of the project.

3.2.1 Regular project team meetings

To keep the entire project team informed on project progress, the **Lead RSE** together with the LA organizes a periodical project meeting. The frequency and format depend on the complexity of the project and size of the project team.

Table 7: Project team meeting overview

Frequency:	Determined by the project team; commonly every two to four weeks.
Participants:	Lead RSE (organizer), LA, RSEs, and relevant members of the LA team.
Objective:	Coordinate work, assess progress, resolve issues, and update planning.
Output:	Agreed actions and relevant decisions recorded in the project log (see Section 3.3.1).
Duration:	Typically 30–60 minutes.
Location:	In person or online, as agreed by the project team.

The agenda of this meeting should include:

- Status update from all stakeholders
- Discussion of scientific progress
- Discussion of technological progress, issues and choices
- Alignment of project progress with the project workplan, and adjustment of the latter, if necessary.

Responsible: **Lead RSE (co-organizer with the LA).**

3.2.2 Internal coordination and escalation

The **Lead RSE** and the responsible **SH** maintain regular communication throughout the project. Internal coordination may take the form of scheduled meetings or ad hoc discussions, depending on the complexity and needs of the project. The **Lead RSE** keeps the responsible **SH** informed of the project's progress and of significant issues that may affect its successful delivery.

Internal coordination may cover topics such as:

- progress against the project workplan and agreed deliverables;
- significant changes to the Technology Plan (Section 2.7);
- project risks, issues, and mitigation actions;
- staffing, resource allocation, and budget monitoring;
- workshops, community engagement, and communication activities;
- opportunities for collaboration with other projects or reuse of software; and
- other matters requiring organizational support or a management decision.

Issues that cannot be resolved within the project team, or that materially affect the project's objectives, budget, timeline, staffing, contractual obligations, or collaboration, should be escalated to the responsible **SH**. Where appropriate, the **SH** may involve other stakeholders, such as **Finance**, Communications, CM, P&CO, or the MT, to support decision-making or resolve the issue.

Responsible: **Lead RSE** and **SH**.

3.2.3 Workshops and community events

Some projects include workshops or community events as part of their agreed project activities. Their purpose is to foster collaboration, engage research communities, disseminate project results, collect user feedback, or support the adoption and sustainability of research software. The organization, funding, and expected outcomes depend on the applicable funding programme and project agreement.

When a workshop is funded through the project budget, the following process is followed:

1. The LA prepares a scientific agenda and a budget proposal using the project templates available in the *Templates projects* folder [4]. The **Lead RSE** supports the preparation of the workshop and contributes to the scientific agenda.
2. The responsible **SH** reviews the proposed workshop and approves the scientific programme.
3. **Finance** reviews the proposed budget and confirms that the planned expenditure complies with the project conditions.
4. **Finance** notifies the LA of the budget approval and initiates the corresponding financial process.
5. After the workshop, the LA submits the final financial documentation, including the corresponding invoices, to **Finance**.
6. **Finance** verifies the final expenses and processes the remaining payment.

The CM may support the project team in designing workshop agendas and engaging the target community to maximize the long-term impact of the workshop outcomes.

eScience Center–Lorentz Center Competition workshops

Projects awarded through the joint eScience Center–Lorentz Center Competition differ from regular project workshops. The workshop proposal is developed and submitted as part of the competition, and the role of the eScience Center project team is defined in the awarded proposal.

The **Lead RSE** and the assigned RSEs support the scientific organizers throughout the preparation, delivery, and follow-up of the workshop in accordance with the agreed project plan. Typical contributions may include scientific and technical advice, workshop organization, development of research software or demonstrators, and the dissemination of workshop outcomes, such as publications, software releases, white papers, or follow-up funding proposals.

Responsible: **Lead RSE** and **LA**.

3.3 Project monitoring

Project monitoring provides the information needed to assess whether the project is progressing according to the agreed objectives and workplan. Throughout the execution phase, the **Lead RSE** monitors project progress, maintains the project log, and, together with the responsible **SH**, ensures that emerging issues are identified early and addressed in a timely manner.

3.3.1 Project log

The project log is the chronological record of significant project events, decisions, and agreements throughout the execution phase. The **Lead RSE** is responsible for maintaining the project log, ensuring that it is shared with the LA, and storing it in the *B – General documents* folder of the corresponding project in the Project portfolio (see [4]).

The project log may be maintained in any suitable shared platform (e.g., a shared document, project wiki, or other collaborative environment), provided that the version stored in the Project portfolio remains up to date. Rather than duplicating information, the project log should provide references to the authoritative project documents, repositories, and other relevant resources. An example project log is provided in Appendix F.

Typical information recorded in the project log includes:

- key meetings, including dates, agreed actions, and important decisions;
- significant changes to project staffing, infrastructure or computing resources;
- workshops, conferences, and other relevant project events;
- links to the RSD project page, repositories, project deliverables, and other supporting documentation;
- updates to the Technology Plan and other project plans; and
- formal project reviews and agreed follow-up actions.

The project log should remain concise and avoid duplicating information already maintained elsewhere. Detailed technical descriptions, planning documents, and review reports should be maintained in their respective project documents, while the project log records when these changes occurred and where the corresponding information can be found.

Responsible: **Lead RSE**.

3.3.2 Progress, hours and budget monitoring

The **Lead RSE** is responsible for monitoring the overall health of the project throughout its execution. This includes tracking progress against the agreed workplan, monitoring the use of project hours, budget, and other project-specific resources, identifying emerging risks, and informing the **SH** and LA when adjustments are required. Such discussions are normally conducted through the internal coordination and escalation process (Section 3.2.2). To support these activities, the **Lead RSE** uses the eScience Center project management systems, including Ganttlic for project planning, Metabase dashboards for project monitoring, and the financial information maintained by **Finance** in AFAS.

The **Lead RSE** routinely monitors:

- progress towards milestones and agreed deliverables;
- project risks, dependencies, and issues affecting delivery.
- expenditure of project hours against the remaining project allocation;
- staffing and resource availability; and
- assess requests for the use of project-specific in-cash resources available for project expenses, such as workshops or open-access publication costs, where applicable;

The information gathered through project monitoring provides the basis for the project review cycle (Section 3.6), during which project progress, resource utilisation, and future annual planning are evaluated.

Project information collected during execution also contributes to the periodic portfolio reviews carried out by the Section Heads and Performance Manager (PerfM) to support resource planning and annual allocations.

Responsible: **Lead RSE, SH and Finance**

Table 8: Project monitoring responsibilities

Stakeholder	Responsibilities	Supporting systems
Lead RSE	• Monitor project progress and deliverables • Monitor project hours and resource allocation • Monitor project-specific cash resources • Identify and communicate risks and deviations • Escalate issues to the SH when required	Ganttlic, Metabase
SH	• Oversee project health • Support corrective actions • Decide on organizational escalations	Ganttlic, Metabase
Finance	• Maintain financial information • Confirm the available in-cash balance and eligibility of proposed expenses • Process approved expenses and payments • Provide updated budget information to the Lead RSE and SH	AFAS, Metabase

3.3.3 Risks, issues and project changes

Projects may encounter scientific, technical, organizational, or resource-related risks during their execution. The **Lead RSE** is responsible for identifying emerging risks and issues at an early stage and discussing them with the LA and project team whenever possible. Significant risks and unresolved issues should be communicated promptly to the responsible **SH** through the internal coordination and escalation process (Section 3.2.2) and are reviewed as part of the project review cycle (Section 3.6).

Projects may require adjustments during their lifetime due to changes in scientific priorities, technology, staffing, available resources, project planning, funding conditions, or other project circumstances. The **Lead RSE** discusses proposed changes with the LA and the responsible **SH**. Changes are handled according to the governance process summarized in Table 9.

Table 9: Project change governance

Type of change	Decision by
Minor adjustments to project planning, milestones or day-to-day activities.	Lead RSE and LA
Changes to project staffing and resource allocation within the approved project budget.	SH
Significant changes affecting project objectives, project budget, continuation or termination.	MT
Changes to externally funded projects.	According to the applicable funding agreement (see Appendix A).

Project-related disagreements should be addressed as early as possible by the project team. Where necessary, the **Lead RSE** escalates the matter to the responsible **SH**. The **SH** may involve other stakeholders or bring the matter to the MT when an organizational decision is required. Issues concerning employment conditions, personal conduct, or workplace behaviour are handled according to the Personnel Handbook [15] and the Code of Conduct [16].

3.4 Project quality and outputs

Throughout project execution, the project team maintains a range of practices and documentation to ensure the quality, sustainability, and long-term value of the project's outputs. These include software engineering best practices, management plans, project documentation, and the registration of project outputs. Together, they support project monitoring, facilitate collaboration, and contribute to the reproducibility and reuse of project results.

3.4.1 Software quality and sustainability

Producing high-quality, sustainable research software is a fundamental objective of the eScience Center. Software developed within projects should follow established software engineering best practices and be designed to maximise reusability, maintainability, reproducibility, and long-term sustainability. Project teams are expected to follow the eScience Center software development guide [14] and the GenAI Policy [17] throughout the software development life cycle.

Code review

The **Lead RSE** is responsible for ensuring that appropriate code review practices are followed throughout the project. Rather than being a single activity, code review is a continuous process integrated into software development. Reviews may be performed by project team members, other RSEs, or with the support of AI-assisted code review tools, where appropriate and in accordance with the eScience Center GenAI Policy.

Code reviews should aim to:

- improve software quality, readability, and maintainability;
- identify defects and potential security or performance issues;
- promote knowledge sharing within the project team;
- encourage alignment with the Technology Plan and eScience Center software development best practices; and
- identify opportunities for software reuse and long-term sustainability.

The Section Head oversees that appropriate software quality practices are followed throughout the project. Where additional expertise is required, the **Lead RSE** is encouraged to seek feedback from the project team, a Consulting RSE, relevant Special Interest Groups (SIGs), or other technical experts within the eScience Center.

Responsible: Lead RSE.

3.4.2 Data and Software Management Plans

Projects awarded through eScience Center calls require a Data Management Plan (DMP [18]) and a Software Management Plan (SMP [19]). These documents describe how project data and research software are managed throughout the project life cycle, including aspects such as documentation, accessibility, sustainability, and long-term preservation. Additional requirements may apply for externally funded projects, depending on the applicable funding programme.

The LA is responsible for the drafting and submission of the DMP and SMP, while the **Lead RSE** supports their maintenance. Both plans are living documents and should be reviewed and updated whenever significant changes affect the project or its technical implementation. The status of the DMP and SMP is reviewed as part of the annual project review (Section 3.6).

Responsible: LA and Lead RSE.

3.4.3 Project documentation

Throughout the project life cycle, the **Lead RSE** and LA maintain a set of documents that support project execution, technical direction, and quality assurance:

- the **Project Log**, recording key project decisions, milestones, and events;
- the **Technology Plan**, describing the project's technical strategy and software engineering approach;
- the **SMP/DMP**, management of research software and data;
- the **Project Status Report**, prepared as part of the annual project review; and
- any additional documentation required by the funding programme or the project agreement.

Together, these documents ensure that project decisions, technical plans, outputs, and progress are properly documented and can be reviewed throughout the project life cycle.

Responsible: Lead RSE and LA.

3.4.4 Project outputs

Projects generate a wide range of scientific and technical outputs throughout their lifetime. These may include research articles, software, datasets, presentations, posters, workshops, tutorials, reports, blog posts, interviews, videos, training materials, and other dissemination activities. The project team is expected to manage these outputs in accordance with the eScience Center's Special Conditions for project funding [6] and the eScience Center GenAI Policy [17].

In particular, project teams are expected to:

- Maximise the scientific impact, visibility, and reuse of project outputs.
- Publish scientific publications as Open Access.
- Develop reusable research software and release it under an Open Source license, preferably Apache License 2.0.
- Register research software and other project outputs in the RSD and assign persistent identifiers (e.g., DOIs via [Zenodo](#)).
- Acknowledge the Netherlands eScience Center and include the project grant identifier in publications and other dissemination activities.
- Ensure appropriate recognition of eScience Center RSE contributions through authorship.
- Record project outputs in the project portfolio.

In addition, the eScience Center promotes the application of the FAIR principles for research data and research software to all project outputs whenever possible [20, 21].

The **Lead RSE** should ensure that the project's RSD page is kept up to date throughout the project lifecycle, including the registration of project outputs, software, publications, and other relevant information. Guidance on maintaining RSD project pages is provided in Appendix E.

When publication fees apply, project teams should first explore Open Access publishing opportunities available through the LA's institution, for example institutional Open Access agreements or library-supported publishing arrangements. Project-specific in-cash resources may also be used to cover publication costs, if available.

For externally funded projects, publications and research software must additionally comply with the Open Access, Open Science, and dissemination requirements of the applicable funding programme (see Appendix A).

Project teams are encouraged to work with the Communications team to maximise the visibility and scientific impact of significant project outcomes through appropriate dissemination activities, such as news articles, blog posts, interviews, videos, and outreach events. Significant project outputs should be recorded in the Project Log and, when appropriate, included in the annual Project Status Report.

Responsible: Project team.

3.5 Knowledge transfer and communication

Knowledge transfer between the eScience Center and the LA's team is a continuous activity throughout the project. The **Lead RSE** ensures that project knowledge, software, documentation, and technical decisions are shared with the project team, enabling researchers, inside and outside the project team, to effectively use, maintain, and further develop the project results beyond the project lifetime.

Lead RSEs are encouraged to actively share project knowledge and outcomes both within and outside the eScience Center. Internally, this includes participating in Special Interest Groups (SIGs), coordinated by the TO, presenting project experiences at Section meetings, and exchanging technical knowledge with colleagues. Externally, project teams are encouraged to disseminate project results through publications, presentations, workshops, software demonstrations, blog posts, interviews, videos, and other outreach activities. Communications and Section Heads support the project team in promoting significant project outcomes through appropriate communication channels, while the **Lead RSE** acts as the primary contact for coordinating these activities.

The project team should ensure that project knowledge is sufficiently documented to support onboarding of new team members and continuity in the event of staffing changes. The Project Log, Technology Plan, Software Management Plan, and other project documentation provide the foundation for this knowledge transfer.

The LA and their team are encouraged to participate in relevant community events and Digital Skills workshops, coordinated by the Training Programme Coordinator (TPC). Where appropriate, project-specific training activities may also be organised with support from the TPC, CMs, and other relevant stakeholders.

Throughout project execution, the project team is encouraged to identify opportunities for software reuse, community engagement, follow-up funding, and collaboration beyond the scope of the current project.

Responsible: **Lead RSE**, with support of the **LA**, **SH**, **TPC**, **CM** and **Communications**.

3.6 Project review

The annual project review provides a structured process for evaluating project progress and determining the continuation of eScience Center support [6]. It aligns project evaluation, portfolio planning, and internal decision-making before the start of each calendar year. The review considers scientific progress, collaboration, project execution, resource utilisation, project risks, and future plans, using the information gathered throughout project monitoring (Section 3.3.2) as the basis for the evaluation.

In addition to assessing the current status of the project, the annual review provides an opportunity to identify opportunities for increasing scientific impact, promoting software reuse and sustainability, strengthening community engagement, and exploring future collaborations or follow-up funding arising from the project.

At the conclusion of the annual project review, each project receives one of the following outcomes:

- **Continue:** Project support continues with the planned allocation of project hours.
- **Adjust:** Project support continues with revised project planning and/or a revised allocation of project hours.
- **Stop:** eScience Center support is concluded.

The annual project review consists of five consecutive stages, from the preparation of the Project Status Report to the communication and implementation of the final decision.

3.6.1 Phase 1 – Initiating the review cycle

To initiate the annual review cycle, the responsible **SH** requests a Project Status Report for each active project. The template can be found in the *Templates projects* folder [4] and provides a concise overview of the project. It typically includes:

- project information and RSD completeness;
- progress against the agreed project workplan;
- project risks, mitigation actions, and planned activities;
- status of project management plans (e.g., Technology Plan, SMP and DMP);
- software, data, publications, outreach activities, and project impact;
- software quality, sustainability, and community engagement; and
- collaboration within the project team and any relevant remarks for the review.

The **Lead RSE** coordinates the preparation of the report with the LA and schedules the review meeting.

Table 10: Initiation of the annual project review

Timing:	Early to mid-September.
Initiated by:	SH
Coordinated by:	Lead RSE
Objective:	Launch the annual project review and collect the information required to evaluate project progress and prepare the review meeting.
Actions:	The Lead RSE sends the Project Status Report template to the LA, requests its submission by mid-October, and schedules the project review meeting in November.
Output:	Project Status Report requested and review meeting preparation initiated.

The LA prepares the Project Status Report with support from the **Lead RSE**, who may assist in completing the report where required. The Secretary may support the **Lead RSE** with the practical organization of the review meeting.

Responsible: SH.

3.6.2 Phase 2 – Internal project assesment

Before the review meeting, the **SH** reviews the submitted reports and identifies projects requiring additional discussion. Where appropriate, additional stakeholders (for example the PO, TO or CM) may be invited to participate in the review meeting.

Responsible: SH.

3.6.3 Phase 3 – Review meeting

The annual project review meeting provides a structured opportunity to evaluate the overall status of the project, discuss matters requiring attention, and prepare the continuation assessment for the next review period. The meeting is based on the submitted Project Status Report and should focus on clarification, discussion, and decisions rather than repeating information already recorded in the report.

The review presentation is prepared jointly by the LA, **Lead RSE**, and **SH** using the project review template available in the *Templates projects* folder [4]. The LA presents the scientific status of the project, the **Lead RSE** presents the

research software and technical progress, and the **SH** summarizes the outcome of the discussion and prepares the preliminary continuation recommendation.

The agenda of the review meeting should include:

- an introduction outlining the purpose and expected outcomes of the meeting;
- project status, including scientific progress, milestones, deliverables, upcoming activities, and scientific risks or issues;
- research software and technical progress, including technical outputs, project management plans, software sustainability, infrastructure requirements, and technical risks;
- a joint review and discussion of collaboration, project risks and bottlenecks, resource utilisation, administrative or contractual matters, and future opportunities; and
- a continuation assessment, including the key conclusions, agreed actions, and preliminary recommendation for project continuation.

Table 11: Annual project review meeting

Timing:	Throughout November.
Organizer:	Lead RSE.
Participants:	Lead RSE , LA, SH , and, where appropriate, additional stakeholders such as the PO, TO, or CM.
Objective:	Evaluate project progress, discuss matters requiring attention, and gather the information needed for the continuation assessment.
Actions:	The LA presents the scientific project status, the Lead RSE presents the research software and technical progress, all participants contribute to the review and discussion, and the SH summarizes the conclusions and agreed actions.
Output:	Agreed actions and a preliminary continuation recommendation recorded in the review materials.
Duration:	Typically 90 minutes; simpler projects may require only 60 minutes.
Location:	In person at the eScience Center or the project location. Online participation may be arranged where appropriate.

Following the review meeting, the **SH** completes the evaluation section of the Project Status Report, incorporating the agreed conclusions, action points, and preliminary continuation recommendation. The completed Project Status Report and the review presentation are then stored in the corresponding project folder within the Project Portfolio. The **SH** finalizes the project evaluation following the MT continuation decision, as described in Phase 4.

Responsible: **Lead RSE.**

3.6.4 Phase 4 – Continuation decision

Following the review meetings, the **SH** consolidates the evaluations for all projects within their section and prepares a continuation recommendation for each project, incorporating any adjustments agreed during the review meetings.

The project evaluations are submitted to the MT for confirmation. Projects progressing as expected are normally approved without further discussion. The MT focuses its deliberations on projects requiring significant changes, major budget reallocations, or where continuation of eScience Center support is uncertain.

The continuation decision forms the basis for the financial processing and communication activities described in Phase 5.

Responsible: SH and MT.

3.6.5 Phase 5 – Financial processing and communication

Following the MT continuation decision, the approved decisions are formally recorded and communicated to **Finance**. **Finance** completes the required administrative processing, updates the relevant project administration systems, and prepares any required continuation or termination letters. The confirmed project status is then communicated to the project team and the LA through the appropriate administrative channels.

Table 12: Administrative processing and communication

Timing:	Early to mid-December.
Responsible:	Finance.
Objective:	Finalize the approved continuation decisions and update the project administration.
Actions:	Process the MT decisions, update AFAS and other administrative systems, prepare any required continuation or termination letters, and communicate the approved outcomes through the established administrative process.
Output:	Updated project administration and communicated continuation decisions.

Responsible: **Finance.**

The execution phase concludes with the annual project review and the corresponding continuation decision. Depending on the outcome, the project either proceeds into a new execution period or enters the closing phase, where the project deliverables, reporting obligations, and administrative activities are formally completed.

4 Project closing

Project closing is the final phase of the project lifecycle. Its purpose is to formally conclude the collaboration, verify the completion of the project objectives and deliverables, document the project outcomes and ensure that all administrative outstanding tasks are completed. Project closing consists of two stages: the closing meeting and the end report.

4.1 Closing meeting

The closing meeting marks the conclusion of the collaboration between the eScience Center and the Lead Applicant. It is typically held as the last regular project meeting 3.2.1 during the last weeks of the project. The meeting provides an opportunity to review the project outcomes, discuss any remaining activities, agree on responsibilities after the project ends, and prepare the end report.

Typical discussion topics include:

- review of the project objectives and deliverables,
- outstanding publications, software releases or other project outputs,
- remaining knowledge transfer activities,
- software sustainability and maintenance plans,
- sharing the end report template, and
- opportunities for future collaboration.

The conclusions and action points are recorded in the project log.

Responsible: Lead RSE.

4.2 End report

The end report documents the outcomes of the project and provides the information required for the formal closure of the project. For internal projects, the end report is written by the LA using the template available in the *Project templates* folder of the project portfolio and previously introduced by the Lead RSE at the closing meeting.

Table 13: End report process overview

Preparation:	The LA completes the end report using the template available in the <i>Project templates</i> folder and submits it within three months after project completion.
Scientific review:	The Lead RSE reviews the scientific content, verifies that all project outputs have been registered in the appropriate systems, and requests revisions from the LA if needed.
Approval:	The responsible SH reviews and approves the end report before it is submitted to Finance for administrative processing.
Financial review:	Finance reviews the financial information, where applicable, and requests corrections if needed.
Formal closure:	Once all reviews have been completed, Finance archives the required documentation and issues the formal project closing letter to the LA, thereby formally concluding the project.

For external projects, the reporting requirements of the funding organisation take precedence (see Appendix A). Whenever possible, the report submitted to the funding organisation is also archived in the project portfolio.

The end report typically includes a public summary, project outcomes, challenges, opportunities, sustainability plans, project deliverables, workshops, and, where applicable, a financial overview. The completed report must be submitted within three months after the end of the project.

After receiving the completed end report, the **Lead RSE** reviews the scientific content and ensures that all project deliverables and outputs have been registered in the appropriate systems. This includes updating the RSD project page, including the public summary and any missing project metadata (see Appendix E), archiving the latest project documentation (e.g., the end report, Technology Plan, DMP and SMP, where applicable) in the project portfolio, and requesting revisions from the LA if necessary. Once the scientific review has been completed, the responsible **SH** approves the report and forwards it to **Finance** for review of the financial information. Communications may use the public summary and project outcomes for dissemination activities.

If the report is approved, **Finance** completes the administrative closure of the project, archives the required documentation in the project portfolio, and issues the formal project closing letter to the LA, thereby formally closing the project and ending the project lifecycle.

Related links and references

- [1] The Netherlands eScience Center. 2015 NLeSC Protocol voor Calls, Subsidietoekenning en Projectmanagement.
- [2] The Netherlands eScience Center. What is a Research Software Engineer? A definition by the Netherlands eScience Center. <https://doi.org/10.5281/zenodo.7994286>, 2023.
- [3] The Netherlands eScience Center. Research Software Engineer at the Netherlands eScience Center: Job Description. <https://doi.org/10.5281/zenodo.7805870>, 2023.
- [4] The Netherlands eScience Center. Project Portfolio (the internal project repository).
- [5] Carlos Martinez-Ortiz, Rena Bakhshi, Yifat Dzigan, Nicolas Renaud, Faruk Diblen, Berend Weel, Maarten van Meersbergen, Niels Drost, Sven van der Burg, and Fakhreh Alidoost. Structured and Unstructured Teams for Research Software Development at The Netherlands eScience Center. *Computing in Science & Engineering*, 24(3):25–32, 2022. doi:10.1109/MCSE.2022.3167448.
- [6] The Netherlands eScience Center. Bijzondere voorwaarden Netherlands eScience Center subsidies. Available via <https://www.esciencecenter.nl/calls-for-proposals/>. Accessed: July 29, 2026.
- [7] J. H. Spaaks. ASReview & HOLM plan. <https://doi.org/10.5281/zenodo.7849050>, April 2023. Zenodo. doi:10.5281/zenodo.7849050.
- [8] The Turing Way Community. The Turing Way: A handbook for reproducible, ethical and collaborative research, April 2025. doi:10.5281/zenodo.3233853.
- [9] Jeff Sutherland and Ken Schwaber. The Scrum Guide. <https://scrumguides.org/index.html>.
- [10] AgileFirst. Shape up: A complete guide to this new development methodology. <https://agilefirst.io/what-is-shape-up/>, 2024.
- [11] OMIMO. micro.p3.express, minimalist project management for micro projects. <https://micro.p3.express/>, 2023.
- [12] Kent Beck and Cynthia Andres. *Extreme Programming Explained: Embrace Change*. Addison-Wesley Professional, 2nd edition, 2004. ISBN: 978-0321278654.
- [13] Greg Wilson, DA Aruliah, CT Brown, NPC Hong, M Davis, RT Guy, SHD Haddock, KD Huff, IM Mitchell, MD Plumbley, et al. Best practices for scientific computing. *PLoS biology*, 12(1):e1001745, 2014. doi:10.1371/journal.pbio.1001745.
- [14] Niels Drost, Jurriaan H. Spaaks, Bouwe Andela, Lourens Veen, Janneke M. van der Zwaan, Stefan Verhoeven, Patrick Bos, Mateusz Kuzak, Ben van Werkhoven, Jisk Attema, Johannes Hidding, Vincent Hees, Carlos Martinez-Ortiz, Hanno Spreeuw, Joris Borgdorff, Katrin Leinweber, Faruk Diblen, Gijs van den Oord, Romulo Goncalves, Arnold Kuzniar, Dafne van Kuppevelt, Berend Weel, Christiaan Meijer, Jason Maassen, Pablo Rodríguez-Sánchez, Tom Klaver, Willem Robert van Hage, Felipe Zapata, and Tom Bakker. *Netherlands eScience Center - Software Development Guide*. Zenodo, September 2020. doi:10.5281/zenodo.4020564.
- [15] The Netherlands eScience Center. Intranet. <http://intranet.esciencecenter.nl/>.
- [16] The Netherlands eScience Center. CAO & Personnel policy. CAO is available via <https://www.wvoi.nl/>.
- [17] Netherlands eScience Center. Policy concerning the use of Generative AI at the Netherlands eScience Center (0.92). <https://doi.org/10.5281/zenodo.10118669>, 2023. Zenodo. doi:10.5281/zenodo.10118669.

- [18] Maria Cruz and Eveline van den Berg. NWO DMP assessment rubric, January 2020. [doi:10.5281/zenodo.3629157](https://doi.org/10.5281/zenodo.3629157).
- [19] Carlos Martinez-Ortiz, Paula Martinez Lavanchy, Laurents Sesink, Brett G. Olivier, James Meakin, Maaïke de Jong, and Maria Cruz. Practical guide to Software Management Plans, October 2022. [doi:10.5281/zenodo.7248877](https://doi.org/10.5281/zenodo.7248877).
- [20] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, and *others*. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3(1):160018, Mar 2016. [doi:10.1038/sdata.2016.18](https://doi.org/10.1038/sdata.2016.18).
- [21] Neil P. Chue Hong, Daniel S. Katz, Michelle Barker, Anna-Lena Lamprecht, Carlos Martinez, Fotis E. Psomopoulos, Jen Harrow, and RDA FAIR4RS WG. FAIR Principles for Research Software (FAIR4RS Principles), June 2022. [doi:10.15497/RDA00068](https://doi.org/10.15497/RDA00068).
- [22] The Netherlands eScience Center. Netherlands eScience Center external projects policy. [tobeadded](#)To be added. Accessed: November 1 2023.

Appendices

A Applying the protocol to external projects

Projects funded by external organisations (e.g., NWO, Horizon Europe, industry, or public-private bilateral partnerships) are subject to contractual obligations defined by the funding agreement, collaboration agreement, or employment contract. Where these obligations differ from the guidance provided in this protocol, the contractual requirements take precedence.

This appendix explains how the Project Management Protocol applies to externally funded projects and highlights the main adaptations required to comply with funding conditions while remaining consistent with the eScience Center's internal project management practices. Further information on the objectives and scope of external projects and the approval process for pursuing externally funded projects is available in the eScience Center's External Projects Policy [22].

A.1 Project initiation

The starting point of an externally funded project is determined by the contractual arrangements with the funding organisation (e.g. consortium agreement or 1-on-1 contract). Depending on the funding scheme, project initiation may consist of a bilateral kick-off meeting with the LA, or of a formal consortium kick-off meeting involving all project partners. Regardless of the external arrangements, the eScience Center follows the initiation activities described in this protocol, unless they conflict with contractual requirements.

Some deviations may include:

- The project start date is defined by the funding agreement.
- Consortium kick-off meetings may replace or complement the project kick-off described in this protocol.
- Additional project documentation (e.g., Grant Agreement, Consortium Agreement, IP Agreement) should be registered together with the project records in the corresponding project portfolio folder.
- A separate Technology Plan may not be required if the project proposal or contract arrangement already defines the project's technical choices and implementation approach.

A.2 Project execution

Unless specified otherwise by the funding agreement or consortium arrangements, the project execution practices described in this protocol remain applicable. This includes regular project and internal coordination meetings, maintenance of the project log and RSD page, recording of project hours, and the software quality and sustainability practices promoted by the eScience Center.

In addition to the above, some adaptations may be required to comply with the contractual obligations of the funding organisation. These may include:

- Project reporting, deliverables, milestones, and review processes are governed by the funding agreement and take precedence over the reporting recommendations described in this protocol. These activities are usually handled and led by the project coordinator of the funding organization.
- Financial information required for external reports should be obtained from **Finance**. **Lead RSE** should keep key project and reporting documentation records in the project portfolio folder.
- DMPs, SMPs and other project management artefacts should follow the format and schedule required by the funding organisation.

-
- Project activities such as dissemination, communication, community building, and similar activities should be carried out in accordance with the contractual requirements.

A.3 Project closing

The project closing activities described in this protocol remain applicable. In addition to the above, the following adaptations may be required to comply with the contractual obligations of the funding organisation:

- The project closing meeting may be organised by the funding organisation or the project coordinator and may replace or complement the closing meeting described in this protocol.
- The final project report is typically coordinated by the funding organisation or project coordinator. The eScience Center should retain a copy of the final report and complete any internal administrative procedures required to close the project.
- Once the final report has been approved, the responsible **SH** informs **Finance** that the project can be administratively closed.
- 1-on-1 service contracts may not require a formal end report. In such cases, project closure follows the contractual acceptance procedure agreed with the customer.

B Applying the protocol to internal projects

C Lead RSE role description

This role description includes guidelines that need to be followed by RSEs fulfilling the role; they will be complemented by protocols.

1. Role	Lead RSE
2. Place in the organization	Project
3. Contacts	SH , project team RSEs, external project partners
4. Purpose	To be responsible for the day-to-day running of a research project at the eScience Center and to act as the main point of contact for the project. Each project has one Lead RSE .
5. Main tasks & responsibilities	• Agreeing with the accountable SH on the division of tasks and the allocation of time within the project • Coordinating day-to-day activities with other RSEs working on the project • Making sure that agreed activities, procedures and targets are carried out and met on time • Monitoring project progress, including project hour expenditure, and regularly reporting progress to the accountable SH • Ensuring that major technological decisions are appropriately reviewed and monitoring their implementation, in consultation with the accountable SH • Ensuring that generalization and re-usability opportunities are considered and implemented from the start of the project, in consultation with the accountable SH • Ensuring the presence of the accountable SH project at meetings where their participation is required (e.g. project kick-off and review meetings) • Solving everyday technical and managerial problems and escalating them to the accountable SH when necessary • Making sure all project outputs are properly released, documented and archived in the designated systems • Ensuring the visibility of the project through project demonstrators, slide decks and other appropriate dissemination activities
6. Competencies	• Leading • Project management • Communicating • Cooperating • Result orientation • Decision making
7. Available resources (budget, hours, training)	In project budget
8. How to get this role	SH assigns Lead RSE based on skills, experience, knowledge, interest and availability

D Project lifecycle checklist

Project Initiation

- ☐ Section Heads: Assign project to a **SH**
- ☐ **Finance**: Register project in AFAS
- ☐ **SH**: Assign **Lead RSE** and eScience project team
- ☐ **Lead RSE**: Create RSD page
- ☐ **SH**: Hold administrative kick-off meeting
- ☐ **Lead RSE**: Organize project kick-off meeting
- ☐ **SH**: Arrange legal agreements
- ☐ **Lead RSE**: Draft technology plan

Project Execution

At project start

- ☐ **Lead RSE**: Update project planning in Ganttlic
- ☐ **Lead RSE**: Organize regular project meetings
- ☐ **Lead RSE**: Organize internal coordination meetings

Throughout the project

- ☐ **Lead RSE**: Maintain project log
- ☐ **Lead RSE**: Monitor project hours and budget
- ☐ **Lead RSE**: Keep Ganttlic planning up to date
- ☐ **Lead RSE**: Review and document risks and issues
- ☐ **Lead RSE**: Register project outputs, update RSD page

Milestones

- ☐ LA & **Lead RSE**: Submit DMP within 3 months of project start
- ☐ **SH**: Organize annual project review

Project Closing

- ☐ LA: Submit end report
- ☐ **Lead RSE** & **SH**: Complete scientific review
- ☐ **Lead RSE**: Update RSD page with final project information
- ☐ **Finance**: Complete financial review
- ☐ **Finance**: Archive final deliverables and documentation
- ☐ **Finance**: Send closing letter and formally close project

E RSD project pages

Each project at the eScience Center has an [RSD page](#).

Project Initiation At the project start, the following data should be added to the RSD project page:

- Grant ID: currently, the Awarding letter reference, for External funders (e.g., EU, NWO) often provide their own ID.
- Funder(s): list of all funding organizations, including the eScience Center.
- Dates: project start and end dates.
- Participating organizations: list of the organizations, including the eScience section(s).
- Research domain: relevant project domain, often from the proposal. Multiple domains can be chosen.
- Description: typically copied from the proposal abstract.
- A cover image: initially a placeholder image provided by the Communication.
- Team and maintainers: the **Lead RSE** and the project team members, including the LAs and their team, with roles and ORCIDs where possible. By default, **Lead RSE** and **SH** are listed as a maintainer, while the inclusion of the LA is optional.
- Relevant eInfrastructures used.
- Relevant keywords and categories.
- eScience funding call name, if applicable.

During the project lifecycle, the **Lead RSE** is responsible for managing page updates.

Project Execution During the project, the following information should be added or updated as needed:

- Repository: URL to the project's GitHub repository (or other version control platform).
- Project Team: List of all project team members, including the LAs and their team, with roles and ORCIDs where possible. The **Lead RSE** should keep this list up to date as team members join or leave the project and use the role field to indicate periods of involvement.
- Related Projects: Links to predecessor or related projects in the RSD, where applicable.
- Project Outputs: Register project outputs as they become available, including:
 - Research outputs: Publications, published datasets, preprints, presentations, posters, reports, websites, notebooks, tutorials, blogs, videos, interviews, news articles, and other dissemination materials. Where available, use persistent identifiers (e.g., publisher DOIs or Zenodo DOIs). Online resources without DOIs may be registered using their URL and a short description.
 - Research software: Reusable software should have its own RSD software page and be linked to the project. Single-use software (e.g., scripts) may also be archived in Zenodo and referenced using a DOI. Versioned software releases archived in Zenodo can be linked when software has been developed across multiple projects.
 - Project documentation: Design documents such as the Technology Plan, SMP, and DMP may be published on Zenodo and referenced using their DOIs.

Project Closing Before the project is closed, the **Lead RSE** should review the RSD project page to ensure that it provides a complete and accurate public record of the project. In particular, verify that:

- Project Description: The project description accurately reflects the final project outcomes.
- Cover Image: The placeholder image has been replaced by an appropriate project image, where available.
- Project Outputs: Publications, datasets, software, and other significant outputs have been registered and linked.
- Related Software: Relevant RSD software pages are linked to the project.
- Digital Infrastructure: The digital infrastructure used during the project has been documented.
- Funding Information: Funding call information and project metadata are complete and up to date.
- Project Team: The final project team and maintainers are correctly listed. The Lead Applicant may also be added as a maintainer and contact person to enable future updates to the project page after the project has ended.
- Testimonials: Testimonials from project partners or the user community have been added, where available.

F Example of the project log

The following example illustrates one possible way of maintaining a project log. The format is intentionally flexible and may be adapted to the needs of the project, provided it offers a clear chronological record of significant project events, decisions, and references to supporting documentation.

Running log for Project XXX

Event	Date	Reference	Notes
Administrative kick-off meeting	2026-03-02	Meeting notes	Project registration completed
Project kick-off meeting	2026-05-09	Slides	Initial workplan agreed
Technology Plan v1	2026-07-18	TechnologyPlan-v1.pdf	Initial version approved
DMP	2026-09-18	Data_Management_Plan.pdf	Initial version approved
Project Review	2026-12-10	Review slides	Continue project
Presentation	2027-01-18	Zenodo DOI	Added to RSD
Software release v1.0	2027-02-05	GitHub release	DOI registered on Zenodo
Research article	2027-03-12	Journal DOI	Added to RSD

2027-03-12 – Research article accepted The publication was accepted, registered in the RSD, and added to the Netherlands eScience Center Zenodo community.

2027-01-24 – SURF infrastructure approved The SURF allocation was approved. The project will use Research Cloud for development and Snellius for production-scale computations.

2027-01-07 – SURF infrastructure request submitted A request for SURF compute and storage resources was submitted based on the agreed Technology Plan.

2026-11-28 – Project meeting In preparation for the project review, the project team met to discuss progress and prepare the presentation slides.

2026-10-25 – Project meeting Meeting with external stakeholders to discuss project progress and next steps.

Agreements:

- Merge the project development branch into the main branch after adding the new tests.
- Prepare a joint presentation for the project review.

2026-06-25 – Project paused The LA was unavailable until 2026-09-01.

: :
: :

2026-05-09 – Project kick-off meeting

Participants – eScience Center: AB (**SH**), AA (**Lead RSE**), AC (RSE), AD (RSE).

Participants – Project team: FA (LA, TU Delft), PA (PhD candidate, TU Delft), RA (postdoctoral researcher, TU Delft).

Agreements:

- Finalize the Technology Plan before 18 September 2026.
- Submit the SURF infrastructure request before the end of September.
- Schedule fortnightly project meetings with the project team.
- Create the project page in the RSD and populate the initial metadata.

Key eScience Roles appearance

Finance, 5, 7, 16–18, 24–26, 29, 30, 33

Lead RSE, 4, 5, 8–23, 25, 26, 29, 32–36

SH, 4, 5, 7–11, 14–18, 21–26, 30, 32–34, 36